Michael Altamirano

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Los Angeles CA

WORK EXPERIENCE

Leidos / FAA Provide engineering project management for the FAA **Dec 2025 – Present**

Electronics Engineer *El Segundo CA*

- Provide electronics engineering support to the Federal Aviation Administration (FAA) Los Angeles Communications Engineering Center, specializing in National Airspace System (NAS) modernization and Legacy Infrastructure Future Technologies (LIFT) program initiatives.
- Design and engineer system integration packages for FAA communications, radar automation, control and monitoring, and weather communication equipment across the Western Service Area
- Lead project engineering, management, and planning for LIFT-related modernization projects induding NEXCOM Radios, Air-to-Ground Protocol Converters, and Airport Cable Loop systems
- Develop comprehensive transition plans, testing protocols, and training programs while coordinating implementation with multiple stakeholders
- Conduct engineering risk assessments and feasibility analyses, providing strategic recommendations for site, facility, and program changes
- Prepare project scope agreements, schedules, and technical documentation supporting critical FAA infrastructure initiatives
- Key Technologies & Systems: National Airspace System (NAS), NEXCOM Radio Systems, Air-to-Ground Protocol Converters, Airport Cable Loop Infrastructure, FAA Communications Equipment

DirecTV prominent provider of digital television entertainment services **Aug 2012 – Feb 2025**

Site Reliability Engineer / Feb 2017 – Feb 2025 *El Segundo CA / Remote*

- Managed and supported an infrastructure of 1,000 servers, applications, and platforms to ensure optimal system performance for VOD and Streaming services.
- Implemented robust queuing solutions using ActiveMQ to streamline messaging and workflow processes within VOD system, enhancing scalability and efficiency.
- Developed/deployed Bash scripts for automating file management and directory structures, utilized Python to build backend services for metadata management, user accounts, and access control, significantly improving system stability.
- Led the deployment of server and blade operations, interpreting complex technical requirements to meet the evolving needs of the video entertainment platform.
- Conducted evaluations of new platforms and executed proofs of concept to identify opportunities for innovation and improvement in system performance.
- Automated responses to key performance metrics and alerts, reducing downtime and enhancing operational efficiency.

Production Engineer / Jan 2015 – Feb 2017 *El Segundo CA*

- Administrated the delivery and management of a VOD library, ensuring seamless service to millions of subscribers.
- Instituted video delivery systems, integrating both legacy satellite and modern IP-based streaming technologies to enhance the VOD platform's reliability and efficiency.
- Coordinated troubleshooting efforts to resolve technical issues such as buffering, error codes, and content availability, significantly improving customer satisfaction and service reliability.
- Collaborated with cross-functional teams, providing technical insights and project management for the successful integration of new technologies, induding video encoding and streaming protocols.

Application Engineer / April 2013 – Jan 2015 *El Segundo CA*

- Operated nationwide Conditional Access Management Center (CAMC) installations for set-top boxes, ensuring secure decryption of premium content for millions of subscribers.
- Piloted a team in implementing robust encryption and smart card upgrades, significantly reducing vulnerabilities to hacking and unauthorized access to paid content.
- Executed and managed set-top box (STB) transactions and load balancing for 21 million subscribers, maintaining service integrity during the DirecTV and AT&T merger integration process.

STB Hardware Engineer / August 2012 – April 2013 *El Segundo CA*

- Developed and implemented test methodologies for transcoder integrated circuits, ensuring optimal performance and compliance with industry standards for video processing systems
- Analyzed MPEG transport streams in transcoder ICs to optimize video delivery quality and system efficiency for set-top box hardware
- Integrated and validated multiple hardware interfaces induding analog video, digital audio, RF front-end systems, FSK communications, IR/RF remote controls, HDMI, USB, and Ethernet connectivity
- Collaborated on hardware design verification and validation processes to ensure seamless integration of video processing components with entertainment delivery systems

International Rectifier/Infineon leader in power management technology **Nov 2000 – Aug 2012**

Device Engineer / October 2009 – August 2012 *El Segundo, CA*

- Ran validation and performance testing for innovative power semiconductor products, contributing to a 20-member team focused on electrical characteristics, thermal performance, and quality control.
- Successfully validated the DirectFET technology, securing a competitive advantage for the company with improved thermal performance and smaller product size, directly addressing critical market needs.

- Worked on the evaluation and validation of the GaN platform, enabling International Rectifier to lead in next-generation power semiconductor technology, thus positioning the company for future growth.
- Conducted comprehensive failure analysis and application-specific testing to ensure product reliability and durability, reinforcing the company's reputation for quality and innovation.

Engineering Technician / November 2000 – October 2009 *El Segundo, CA*

- Conducted comprehensive performance testing and competitive analysis of power semiconductor devices, induding MOSFETs, IGBTs, and diodes, to evaluate electrical characteristics and market positioning
- Designed, modified, and constructed specialized test stations and equipment to support product validation and quality assurance protocols for power management components
- Performed systematic troubleshooting and defect analysis on semiconductor devices, identifying failure modes and implementing corrective measures to improve product reliability
- Executed extreme condition testing procedures, modifying devices and test parameters to evaluate performance under stress conditions and validate design specifications
- Supervised second shift operations and personnel, preparing detailed work summary reports and coordinating testing activities to maintain production schedules and quality objectives

Arroyo Optics, manufacturer of wavelength management solutions for telecoms equipment **1999 – 2000**

Fiber / Chamber Technician *Santa Monica, CA*

- Conducted comprehensive testing and validation of fiber optic couplers to ensure optimal performance and compliance with telecommunications industry standards.
- Executed precision fiber cable splicing operations and maintained calibrated splicing equipment
- Performed detailed analysis and calculations of test results, maintaining accurate documentation and database records
- Operated and maintained sophisticated fiber optic test equipment, ensuring optimal functionality and measurement
- Executed laser welding operations and performed precision tuning of fiber couplers to meet stringent performance

Ortel Corp, packaged lasers and photodiodes coupled to optical fiber, transmitters, and receivers **1997 – 1999**

Fiber / RF Technician *Alhambra, CA*

- Performed comprehensive RF testing induding frequency response analysis, return loss measurements, and final product validation to ensure compliance with telecommunications standards
- Conducted specialized testing procedures induding DC bias analysis, environmental burn-in testing, and characterization
- Executed EPROM programming, component screening, and systematic testing protocols to validate product functionality
- Established and configured test stations for optimal measurement accuracy and efficiency, supporting high-volume production testing
- Collaborated dosely with engineering teams to troubleshoot pending products and communicate technical findings

Allied Signal / JPL, contractor to monitor a project the antenna JPL was building for NASA ` **1996 – 1997**

Computer Technician *Pasadena, CA*

- Maintained critical computer systems and equipment for NASA's Deep Space Network (DSN), ensuring operational continuity for deep space mission communications and data transmission
- Constructed custom adapters and hardware interfaces to support DSN equipment integration and mission-specific requirements
- Executed equipment replacement and upgrade procedures for DSN infrastructure, maintaining system reliability and performance
- Supported ISO 9001 quality management initiatives through documentation, process compliance, and quality assurance activities
- Managed comprehensive inventory control systems for specialized tools and equipment, ensuring asset tracking and availability for critical space communication operations

EDUCATION

California State University, Long Beach	Master of Science in Electrical Engineering	**April 2013**
Keller Graduate School of Management	Master of Business Administration	**Nov 2011**
DeVry University, Long Beach	Bachelors of Science in Computer Engineering Technology	**Feb 2010**
DeVry University, Long Beach	Bachelors of Science Electronics Engineering Technology	**Oct 2009**
Don Bosco Tech Institute, Rosemead CA	Associate in Science in Electronics and Computer Technology	**May 1998**

CERTIFICATIONS, SKILLS & INTERESTS

- **Technologies:** C++, Java, Assembly, Basic, SQL, Linux, UNIX, MATLAB, LabVIEW, VHDL, Verilog.
- **Hardware & Test Equipment:** Oscilloscopes, Spectrum Analyzers, Signal Generators, Fiber Splicers, Laser Welders, Partide Counters, RF/Analog/Digital Test Equipment
- **Skills:** Structured Programming, Data Structures, Database Systems, Signal Processing, RF Communications.
- **Certificates:**
  - Kester Lead Free Hand Soldering and Assembly Course — May 2004
  - Microsoft Certified Professional (MCP) — June 2003
  - Adcon Technical Institute A+, MCSE Windows 2000, & CCNA — June 2003
  - CompTIA Certified in A+ Computer Troubleshooting — October 2002
  - NASA and JPL Certified in Hand Soldering — August 1997
  - NASA and JPL Certified in Wire Wrap — August 1997
  - NASA and JPL Certified in ESD Awareness — August 1997